第 14 章　二重积分

1.【答案】 $\ln 2 \cdot \ln(1+\sqrt{2})$

【解析】
$$\lim_{n\to\infty}\sum_{i=1}^{n}\sum_{j=1}^{n}\frac{1}{(n+i)\sqrt{n^2+j^2}}=\lim_{n\to\infty}\sum_{i=1}^{n}\sum_{j=1}^{n}\frac{1}{\left(1+\frac{i}{n}\right)\sqrt{1+\left(\frac{j}{n}\right)^2}}\cdot\frac{1}{n^2}$$

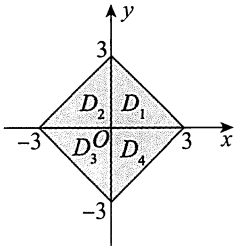
$$=\iint_D\frac{1}{(1+x)\sqrt{1+y^2}}\,\mathrm{d}x\mathrm{d}y=\int_0^1\frac{1}{1+x}\,\mathrm{d}x\int_0^1\frac{1}{\sqrt{1+y^2}}\,\mathrm{d}y$$

$$=\ln(1+x)\Big|_0^1\cdot\ln\left(y+\sqrt{1+y^2}\right)\Big|_0^1=\ln 2\cdot\ln(1+\sqrt{2}),$$

其中 $D=\{(x,y)\,|\,0\leqslant x\leqslant 1,0\leqslant y\leqslant 1\}$.

2.【答案】 (D)

【解析】 如图所示，由轮换对称性知，$I_1=I_3=0$.



在 D_2 上，$-\pi<-3\leqslant x-y<0$；在 D_4 上，$0<x-y\leqslant 3<\pi$.

故 $I_2<0$，$I_4>0$.

3.【答案】 (B)

【解析】 因为当 $x>1$ 时，$\ln x<x-1$，所以在 D 上，当 $x^2+y^2\neq 1$ 时，$\ln(x^2+y^2)<x^2+y^2-1$，从而
$$M=\iint_D\ln(x^2+y^2)\mathrm{d}x\mathrm{d}y<\iint_D(x^2+y^2-1)\mathrm{d}x\mathrm{d}y=P.$$

又因为在 D 上，$1\leqslant x^2+y^2\leqslant 2$，所以在 D 上，

$$0<\ln(x^2+y^2)<1(x^2+y^2\neq 1),\ \ln(x^2+y^2)>\left[\ln(x^2+y^2)\right]^2(x^2+y^2\neq 1),$$

从而
$$N=\iint_D\left[\ln(x^2+y^2)\right]^2\mathrm{d}x\mathrm{d}y<\iint_D\ln(x^2+y^2)\mathrm{d}x\mathrm{d}y=M.$$

于是，$N<M<P$.

4.【解析】 由题意，$\bar{f}=\int_0^1 f(x)\mathrm{d}x=\frac{1}{2}$，又 $f(x)=1-a\int_1^x f(y)f(y-x)\mathrm{d}y$，两边关于 x 作 $[0,1]$ 上的积分，有
$$\int_0^1 f(x)\mathrm{d}x=1-a\int_0^1\mathrm{d}x\int_1^x f(y)f(y-x)\mathrm{d}y=1+a\int_0^1\mathrm{d}x\int_x^1 f(y)f(y-x)\mathrm{d}y$$

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$$=1+a\int_0^1 \mathrm{d}y\int_0^y f(y)f(y-x)\mathrm{d}x=1+a\int_0^1 f(y)\mathrm{d}y\int_0^y f(y-x)\mathrm{d}x.$$

$$\int_0^y f(y-x)\mathrm{d}x \xlongequal{令y-x=t} \int_y^0 f(t)(-\mathrm{d}t)=\int_0^y f(t)\mathrm{d}t,$$

故
$$\int_0^1 f(x)\mathrm{d}x=1+a\int_0^1 f(y)\mathrm{d}y\int_0^y f(t)\mathrm{d}t=1+a\int_0^1\left[\int_0^y f(t)\mathrm{d}t\right]\mathrm{d}\left[\int_0^y f(t)\mathrm{d}t\right]$$

$$=1+\frac{a}{2}\left[\int_0^y f(t)\mathrm{d}t\right]^2\Bigg|_{y=0}^{y=1}=1+\frac{a}{2}\left[\int_0^1 f(t)\mathrm{d}t\right]^2,$$

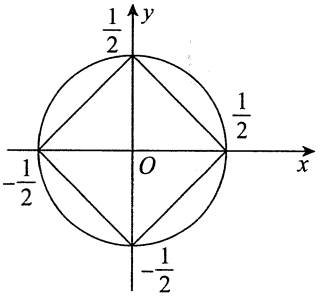
即 $\bar{f}=1+\frac{a}{2}(\bar{f})^2$，也即 $\frac{1}{2}=1+\frac{a}{8}$，得 $a=-4$.

5.【答案】(D)

【解析】 $I_1=\iint\limits_{\substack{0\leqslant x\leqslant 1\\0\leqslant y\leqslant 1}}(\sin x^2+\cos x^2)\,\mathrm{d}\sigma=\sqrt{2}\iint\limits_{\substack{0\leqslant x\leqslant 1\\0\leqslant y\leqslant 1}}\sin\left(x^2+\frac{\pi}{4}\right)\mathrm{d}\sigma.$

由于 $0\leqslant x^2\leqslant 1$，则 $\frac{\pi}{4}\leqslant x^2+\frac{\pi}{4}\leqslant\frac{\pi}{4}+1$，故 $\frac{\sqrt{2}}{2}\leqslant\sin\left(x^2+\frac{\pi}{4}\right)\leqslant 1$，于是 $1\leqslant I_1\leqslant\sqrt{2}$.

令 $D=\left\{(x,y)\middle|x^2+y^2\leqslant\left(\frac{1}{2}\right)^2\right\}$（见图），则



$$\sqrt{x^2+y^2}\leqslant\frac{1}{2},\ \sqrt{x^2+y^2}+\frac{1}{2}\leqslant 1,\ \ln\left(\sqrt{x^2+y^2}+\frac{1}{2}\right)\leqslant 0,$$

故 $I_2\leqslant 2\iint\limits_{|x|+|y|\leqslant\frac{1}{2}}\mathrm{d}\sigma=2\cdot\frac{1}{2}=1.$

综上，$I_2\leqslant 1\leqslant I_1$. 选(D).

6.【答案】 $\frac{5}{16}\pi$

【解析】积分区域如图所示，故

$$\int_{-1}^0\mathrm{d}x\int_{\left(1-x^{\frac{2}{3}}\right)^{\frac{3}{2}}}^{(1-x^2)^{\frac{1}{2}}}(1-\sin x\cos y)\mathrm{d}y+\int_0^1\mathrm{d}x\int_{\left(1-x^{\frac{2}{3}}\right)^{\frac{3}{2}}}^{(1-x^2)^{\frac{1}{2}}}(1-\sin x\cos y)\mathrm{d}y$$

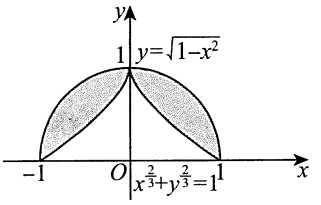
$$=\int_{-1}^1\mathrm{d}x\int_{\left(1-x^{\frac{2}{3}}\right)^{\frac{3}{2}}}^{(1-x^2)^{\frac{1}{2}}}(1-\sin x\cos y)\mathrm{d}y=2\int_0^1\mathrm{d}x\int_{\left(1-x^{\frac{2}{3}}\right)^{\frac{3}{2}}}^{(1-x^2)^{\frac{1}{2}}}\mathrm{d}y$$

$$=2\int_0^1\left[(1-x^2)^{\frac{1}{2}}-\left(1-x^{\frac{2}{3}}\right)^{\frac{3}{2}}\right]\mathrm{d}x$$

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$$=2\int_0^{\frac{\pi}{2}}\cos^2 t\mathrm{d}t-2\int_0^{\frac{\pi}{2}}\cos^3 t\cdot 3\sin^2 t\cdot\cos t\mathrm{d}t$$

$$=2\int_0^{\frac{\pi}{2}}\left[\cos^2 t-3\cos^4 t(1-\cos^2 t)\right]\mathrm{d}t$$



$$=2\int_0^{\frac{\pi}{2}}(\cos^2 t-3\cos^4 t+3\cos^6 t)\mathrm{d}t$$

$$=2\cdot\frac{1}{2}\cdot\frac{\pi}{2}-6\cdot\frac{3}{4}\cdot\frac{1}{2}\cdot\frac{\pi}{2}+6\cdot\frac{5}{6}\cdot\frac{3}{4}\cdot\frac{1}{2}\cdot\frac{\pi}{2}$$

$$=\frac{\pi}{2}-\frac{9}{8}\pi+\frac{15}{16}\pi=\frac{5}{16}\pi.$$

7.【解析】方法一 在极坐标系中，区域D可表示为$\left\{(r,\theta)\middle|0\leqslant r\leqslant 1,0\leqslant\theta\leqslant\frac{\pi}{4}\right\}$，所以

$$\iint_D \mathrm{e}^{(x+y)^2}(x^2-y^2)\mathrm{d}x\mathrm{d}y=\int_0^{\frac{\pi}{4}}\mathrm{d}\theta\int_0^1 \mathrm{e}^{r^2(\sin\theta+\cos\theta)^2}r^3(\cos^2\theta-\sin^2\theta)\mathrm{d}r$$

$$=\int_0^1\mathrm{d}r\int_0^{\frac{\pi}{4}}\mathrm{e}^{r^2(\sin\theta+\cos\theta)^2}r^3(\cos^2\theta-\sin^2\theta)\mathrm{d}\theta=\frac{1}{2}\int_0^1 r\mathrm{d}r\int_0^{\frac{\pi}{4}}\mathrm{e}^{r^2(\sin\theta+\cos\theta)^2}\mathrm{d}\left[r^2(\sin\theta+\cos\theta)^2\right]$$

$$=\frac{1}{2}\int_0^1 r\left[\mathrm{e}^{r^2(\sin\theta+\cos\theta)^2}\Big|_{\theta=0}^{\theta=\frac{\pi}{4}}\right]\mathrm{d}r=\frac{1}{2}\int_0^1 r(\mathrm{e}^{2r^2}-\mathrm{e}^{r^2})\mathrm{d}r=\frac{(\mathrm{e}-1)^2}{8}.$$

方法二 令$\begin{cases}u=x-y,\\ v=x+y,\end{cases}$ $D_{uv}=\left\{(u,v)\middle|u^2+v^2\leqslant 2,v\geqslant u\geqslant 0\right\}$.

因为$\begin{cases}x=\dfrac{u+v}{2},\\ y=\dfrac{v-u}{2},\end{cases}(u,v)\in D_{uv}$，$\left\|\begin{matrix}\dfrac{\partial x}{\partial u} & \dfrac{\partial x}{\partial v}\\ \dfrac{\partial y}{\partial u} & \dfrac{\partial y}{\partial v}\end{matrix}\right\|=\frac{1}{2}$，所以

$$\iint_D \mathrm{e}^{(x+y)^2}(x^2-y^2)\mathrm{d}x\mathrm{d}y=\iint_{D_{uv}}\frac{1}{2}\mathrm{e}^{v^2}uv\mathrm{d}u\mathrm{d}v=\frac{1}{2}\int_0^1 u\mathrm{d}u\int_u^{\sqrt{2-u^2}}\mathrm{e}^{v^2}v\mathrm{d}v$$

$$=\frac{1}{4}\int_0^1(\mathrm{e}^{2-u^2}-\mathrm{e}^{u^2})u\mathrm{d}u=\frac{(\mathrm{e}-1)^2}{8}.$$

8.【答案】0

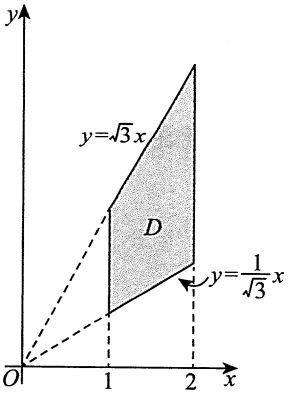
【解析】令$\begin{cases}x=\dfrac{1}{2}r\cos\theta,\\ y=r\sin\theta,\end{cases}$则

$$I=\int_0^{\frac{\pi}{2}}\mathrm{d}\theta\int_0^1(1-r^2-2r^2\cos^2\theta)\frac{r}{2}\mathrm{d}r$$

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$$=\frac{1}{2}\int_0^{\frac{\pi}{2}}\mathrm{d}\theta\int_0^1(1-r^2)r\mathrm{d}r-\int_0^{\frac{\pi}{2}}\cos^2\theta\mathrm{d}\theta\int_0^1 r^3\mathrm{d}r=\frac{\pi}{16}-\frac{\pi}{16}=0.$$

9.【解析】先画出积分区域 D，如图所示.



方法一　用直角坐标.

$$I=\int_1^2\mathrm{d}x\int_{\frac{1}{\sqrt{3}}x}^{\sqrt{3}x}y\mathrm{e}^{\frac{y}{x}}\mathrm{d}y,$$

对内层积分作积分变量代换，令 $u=\dfrac{y}{x}$（视 x 为常量），当 $y=\dfrac{1}{\sqrt{3}}x$ 时，$u=\dfrac{1}{\sqrt{3}}$；当 $y=\sqrt{3}x$ 时，$u=\sqrt{3}$，得

$$I=\int_1^2\mathrm{d}x\int_{\frac{1}{\sqrt{3}}}^{\sqrt{3}}x^2u\mathrm{e}^u\mathrm{d}u=\int_1^2x^2\mathrm{d}x\int_{\frac{1}{\sqrt{3}}}^{\sqrt{3}}u\mathrm{e}^u\mathrm{d}u$$

$$=\frac{7}{3}(u\mathrm{e}^u-\mathrm{e}^u)\Big|_{\frac{1}{\sqrt{3}}}^{\sqrt{3}}=\frac{7}{3}\left(\sqrt{3}\mathrm{e}^{\sqrt{3}}-\mathrm{e}^{\sqrt{3}}-\frac{1}{\sqrt{3}}\mathrm{e}^{\frac{1}{\sqrt{3}}}+\mathrm{e}^{\frac{1}{\sqrt{3}}}\right)$$

$$=\frac{7}{3}(\sqrt{3}-1)\left(\mathrm{e}^{\sqrt{3}}+\frac{1}{\sqrt{3}}\mathrm{e}^{\frac{1}{\sqrt{3}}}\right).$$

方法二　用极坐标.先将 D 化成极坐标形式，为 $D=\left\{(r,\theta)\middle|\dfrac{1}{\cos\theta}\leqslant r\leqslant\dfrac{2}{\cos\theta},\dfrac{\pi}{6}\leqslant\theta\leqslant\dfrac{\pi}{3}\right\}$，所以

$$I=\int_{\frac{\pi}{6}}^{\frac{\pi}{3}}\mathrm{d}\theta\int_{\frac{1}{\cos\theta}}^{\frac{2}{\cos\theta}}r\sin\theta\cdot\mathrm{e}^{\tan\theta}r\mathrm{d}r=\frac{1}{3}\int_{\frac{\pi}{6}}^{\frac{\pi}{3}}\frac{\sin\theta}{\cos^3\theta}\cdot(8-1)\cdot\mathrm{e}^{\tan\theta}\mathrm{d}\theta.$$

令 $u=\tan\theta$，得

$$I=\frac{7}{3}\int_{\frac{1}{\sqrt{3}}}^{\sqrt{3}}u\mathrm{e}^u\mathrm{d}u=\frac{7}{3}(u\mathrm{e}^u-\mathrm{e}^u)\Big|_{\frac{1}{\sqrt{3}}}^{\sqrt{3}}=\frac{7}{3}\left(\sqrt{3}\mathrm{e}^{\sqrt{3}}-\mathrm{e}^{\sqrt{3}}-\frac{1}{\sqrt{3}}\mathrm{e}^{\frac{1}{\sqrt{3}}}+\mathrm{e}^{\frac{1}{\sqrt{3}}}\right)$$

$$=\frac{7}{3}(\sqrt{3}-1)\left(\mathrm{e}^{\sqrt{3}}+\frac{1}{\sqrt{3}}\mathrm{e}^{\frac{1}{\sqrt{3}}}\right).$$

10.【答案】$\ln\cos 1$

【解析】

$$\int_0^1\mathrm{d}x\int_1^x\frac{\tan y}{y}\mathrm{d}y=-\int_0^1\frac{\tan y}{y}\mathrm{d}y\int_0^y\mathrm{d}x$$

$$=-\int_0^1\frac{\tan y}{y}\cdot y\mathrm{d}y=-\int_0^1\frac{\sin y}{\cos y}\mathrm{d}y=\ln\cos 1.$$

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11.【答案】 $\frac{5}{4}\pi$

【解析】 $\iint\limits_D (x-2y)^2 \mathrm{d}x\mathrm{d}y = \iint\limits_D (x^2+4y^2-4xy)\mathrm{d}x\mathrm{d}y$.

由于在区域 D 上，$4xy$ 为 x 的奇函数，区域 D 关于 y 轴对称，因此 $\iint\limits_D (-4xy)\mathrm{d}x\mathrm{d}y = 0$. 所以

$$\iint\limits_D (x-2y)^2 \mathrm{d}x\mathrm{d}y = \iint\limits_D (x^2+4y^2)\mathrm{d}x\mathrm{d}y .$$

由于区域 D 中交换 x 与 y 的位置 D 不变，因此 $I = \iint\limits_D (x^2+4y^2)\mathrm{d}x\mathrm{d}y = \iint\limits_D (y^2+4x^2)\mathrm{d}x\mathrm{d}y$，即

$$2I = \iint\limits_D \left[(x^2+4y^2)+(y^2+4x^2)\right]\mathrm{d}x\mathrm{d}y$$

$$= 5\iint\limits_D (x^2+y^2)\mathrm{d}x\mathrm{d}y = 5\int_0^{2\pi}\mathrm{d}\theta\int_0^1 r^2 \cdot r\mathrm{d}r = \frac{5}{4}\cdot 2\pi = \frac{5}{2}\pi .$$

故 $I = \frac{5}{4}\pi$.

【注】本题也可以不用如上技巧，而直接用极坐标求解. 先用极坐标，再用华里士公式，有

$$\iint\limits_D (x-2y)^2\mathrm{d}x\mathrm{d}y = \int_0^{2\pi}\mathrm{d}\theta\int_0^1 (r\cos\theta - 2r\sin\theta)^2 r\mathrm{d}r$$

$$= \int_0^{2\pi}(\cos^2\theta - 4\cos\theta\sin\theta + 4\sin^2\theta)\mathrm{d}\theta\int_0^1 r^3\mathrm{d}r$$

$$= \frac{1}{4}\int_0^{2\pi}(\cos^2\theta - 4\cos\theta\sin\theta + 4\sin^2\theta)\mathrm{d}\theta$$

$$= \frac{1}{4}\left(4\cdot\frac{1}{2}\cdot\frac{\pi}{2} - \frac{4}{2}\cdot 0 + 4\cdot 4\cdot\frac{1}{2}\cdot\frac{\pi}{2}\right)$$

$$= \frac{5}{4}\pi .$$

12.【解析】 $I = \iint\limits_D \left[(x-1)^2+(2y+3)^2\right]\mathrm{d}x\mathrm{d}y$

$$= \iint\limits_D (x^2+4y^2)\,\mathrm{d}x\mathrm{d}y + \iint\limits_D (-2x+12y)\,\mathrm{d}x\mathrm{d}y + 10\iint\limits_D \mathrm{d}x\mathrm{d}y ,$$

其中 $I_2 = \iint\limits_D (-2x+12y)\mathrm{d}x\mathrm{d}y = 0$（奇函数在对称区域上的积分性质），

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$$I_3=10\iint_D \mathrm{d}x\mathrm{d}y=10ab\pi\text{（椭圆的面积公式）}.$$

$$I_1=\iint_D (x^2+4y^2)\mathrm{d}x\mathrm{d}y=\iint_D x^2\mathrm{d}x\mathrm{d}y+4\iint_D y^2\mathrm{d}x\mathrm{d}y,$$

$$\iint_D x^2\mathrm{d}x\mathrm{d}y=4\int_0^a \mathrm{d}x\int_0^{\frac{b}{a}\sqrt{a^2-x^2}} x^2\mathrm{d}y$$

$$=\frac{4b}{a}\int_0^a x^2\sqrt{a^2-x^2}\,\mathrm{d}x \xlongequal{令x=a\sin t} 4a^3b\int_0^{\frac{\pi}{2}}\sin^2t\cos^2t\mathrm{d}t$$

$$=4a^3b\left(\int_0^{\frac{\pi}{2}}\sin^2t\mathrm{d}t-\int_0^{\frac{\pi}{2}}\sin^4t\mathrm{d}t\right)$$

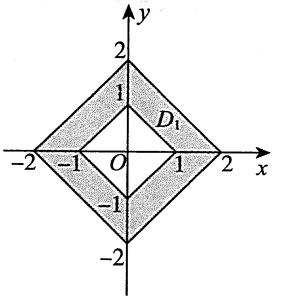
$$=4a^3b\left(\frac{1}{2}\cdot\frac{\pi}{2}-\frac{3}{4}\cdot\frac{1}{2}\cdot\frac{\pi}{2}\right)=\frac{\pi}{4}a^3b.$$

同理，$4\iint_D y^2\mathrm{d}x\mathrm{d}y=\pi ab^3$，则 $I_1=\frac{\pi}{4}a^3b+\pi ab^3$，所以 $I=10\pi ab+\frac{\pi}{4}a^3b+\pi ab^3$.

13. 【解析】设 D_1 是 D 在第一象限的部分，如图所示，则由对称性，得

$$\iint_D \frac{x^2+y^2}{|x|+|y|}\mathrm{d}x\mathrm{d}y=4\iint_{D_1}\frac{x^2+y^2}{x+y}\mathrm{d}x\mathrm{d}y=4\iint_{D_1}\frac{r^2}{\cos\theta+\sin\theta}\mathrm{d}r\mathrm{d}\theta$$

$$=4\int_0^{\frac{\pi}{2}}\mathrm{d}\theta\int_{\frac{1}{\cos\theta+\sin\theta}}^{\frac{2}{\cos\theta+\sin\theta}}\frac{r^2}{\cos\theta+\sin\theta}\mathrm{d}r$$



$$=\frac{28}{3}\int_0^{\frac{\pi}{2}}\frac{1}{(\cos\theta+\sin\theta)^4}\mathrm{d}\theta$$

$$=\frac{7}{3}\int_0^{\frac{\pi}{2}}\frac{1}{\sin^4\left(\theta+\frac{\pi}{4}\right)}\mathrm{d}\theta$$

$$=\frac{7}{3}\int_0^{\frac{\pi}{2}}\csc^4\left(\theta+\frac{\pi}{4}\right)\mathrm{d}\theta$$

$$=-\frac{7}{3}\int_0^{\frac{\pi}{2}}\left[\cot^2\left(\theta+\frac{\pi}{4}\right)+1\right]\mathrm{d}\left[\cot\left(\theta+\frac{\pi}{4}\right)\right]$$

$$=-\frac{7}{3}\left[\frac{1}{3}\cot^3\left(\theta+\frac{\pi}{4}\right)+\cot\left(\theta+\frac{\pi}{4}\right)\right]\Bigg|_0^{\frac{\pi}{2}}=\frac{56}{9}.$$

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14.【答案】 $\frac{1}{2}$

【解析】 交换积分次序，得 $\int_{-1}^{1}\mathrm{d}y\int_{-1}^{y}y\sqrt{1+x^2-y^2}\,\mathrm{d}x=\int_{-1}^{1}\mathrm{d}x\int_{x}^{1}y\sqrt{1+x^2-y^2}\,\mathrm{d}y$.

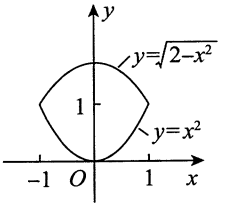
又因为 $\int_{x}^{1}y\sqrt{1+x^2-y^2}\,\mathrm{d}y=-\frac{1}{3}(1+x^2-y^2)^{\frac{3}{2}}\Big|_{x}^{1}=\frac{1}{3}(1-|x|^3)$，所以

$$\text{原式}=\int_{-1}^{1}\mathrm{d}x\int_{x}^{1}y\sqrt{1+x^2-y^2}\,\mathrm{d}y=\frac{1}{3}\int_{-1}^{1}(1-|x|^3)\mathrm{d}x$$

$$=\frac{2}{3}\int_{0}^{1}(1-x^3)\mathrm{d}x=\frac{2}{3}\times\frac{3}{4}=\frac{1}{2}.$$

15.【答案】 $\frac{22}{15}$

【解析】 积分区域如图所示，则

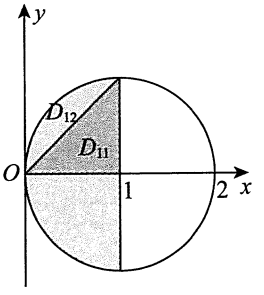


$$\int_{-1}^{1}\mathrm{d}x\int_{x^2}^{\sqrt{2-x^2}}(x+1)y\mathrm{d}y\overset{\text{对称性}}{=\!=\!=}\int_{-1}^{1}\mathrm{d}x\int_{x^2}^{\sqrt{2-x^2}}y\mathrm{d}y$$

$$=2\int_{0}^{1}\mathrm{d}x\int_{x^2}^{\sqrt{2-x^2}}y\mathrm{d}y=\int_{0}^{1}\left(y^2\Big|_{x^2}^{\sqrt{2-x^2}}\right)\mathrm{d}x$$

$$=\int_{0}^{1}(2-x^2-x^4)\mathrm{d}x=\left(2x-\frac{1}{3}x^3-\frac{1}{5}x^5\right)\Big|_{0}^{1}=\frac{22}{15}.$$

16.【解析】 积分区域 D 关于 x 轴对称，记 D_1 为 D 在 x 轴上方的部分，用直线 $y=x$ 将 D_1 分割为 D_{11} 与 D_{12} 两部分，如图所示，则



$$\iint_{D}\big||x-|y||\big|\,\mathrm{d}\sigma=2\iint_{D_1}|x-y|\,\mathrm{d}\sigma$$

$$=2\iint_{D_{11}}(x-y)\mathrm{d}\sigma-2\iint_{D_{12}}(x-y)\mathrm{d}\sigma$$

$$=2\int_{0}^{1}\mathrm{d}x\int_{0}^{x}(x-y)\mathrm{d}y-2\int_{\frac{\pi}{4}}^{\frac{\pi}{2}}\mathrm{d}\theta\int_{0}^{2\cos\theta}r^2(\cos\theta-\sin\theta)\mathrm{d}r$$

$$=\int_{0}^{1}x^2\mathrm{d}x-\frac{16}{3}\int_{\frac{\pi}{4}}^{\frac{\pi}{2}}\cos^3\theta(\cos\theta-\sin\theta)\mathrm{d}\theta$$

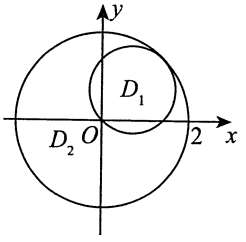
$$=\frac{1}{3}-\frac{4}{3}\int_{\frac{\pi}{4}}^{\frac{\pi}{2}}\left(\frac{3}{2}+2\cos 2\theta+\frac{1}{2}\cos 4\theta\right)\mathrm{d}\theta-\frac{16}{3}\int_{\frac{\pi}{4}}^{\frac{\pi}{2}}\cos^3\theta\mathrm{d}(\cos\theta)$$

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$$=\frac{1}{3}-\frac{4}{3}\left(\frac{3}{2}\theta+\sin 2\theta+\frac{1}{8}\sin 4\theta\right)\Big|_{\frac{\pi}{4}}^{\frac{\pi}{2}}-\frac{4}{3}\cos^4\theta\Big|_{\frac{\pi}{4}}^{\frac{\pi}{2}}$$

$$=\frac{1}{3}-\frac{4}{3}\left(\frac{3\pi}{8}-1\right)+\frac{1}{3}=2-\frac{\pi}{2}.$$

17.【解析】用曲线$\left(x-\frac{1}{\sqrt{2}}\right)^2+\left(y-\frac{1}{\sqrt{2}}\right)^2=1$将$D$划分为$D_1$，$D_2$两部分，如图所示，则有



$$\iint_D\left|x^2+y^2-\sqrt{2}(x+y)\right|\mathrm{d}x\mathrm{d}y$$

$$=-\iint_{D_1}\left[x^2+y^2-\sqrt{2}(x+y)\right]\mathrm{d}x\mathrm{d}y+\iint_{D_2}\left[x^2+y^2-\sqrt{2}(x+y)\right]\mathrm{d}x\mathrm{d}y$$

$$=-\iint_{D_1}\left[x^2+y^2-\sqrt{2}(x+y)\right]\mathrm{d}x\mathrm{d}y+\iint_{D-D_1}\left[x^2+y^2-\sqrt{2}(x+y)\right]\mathrm{d}x\mathrm{d}y$$

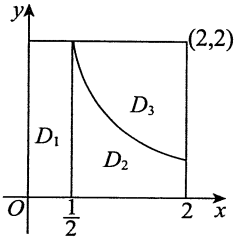
$$=-2\iint_{D_1}\left[x^2+y^2-\sqrt{2}(x+y)\right]\mathrm{d}x\mathrm{d}y+\iint_{D}\left[x^2+y^2-\sqrt{2}(x+y)\right]\mathrm{d}x\mathrm{d}y$$

$$=-4\int_{-\frac{\pi}{4}}^{\frac{\pi}{4}}\mathrm{d}\theta\int_0^{\sqrt{2}(\cos\theta+\sin\theta)}\left[r^3-\sqrt{2}r^2(\cos\theta+\sin\theta)\right]\mathrm{d}r+\iint_D(x^2+y^2)\mathrm{d}x\mathrm{d}y$$

$$=\frac{4}{3}\int_{-\frac{\pi}{4}}^{\frac{\pi}{4}}(\cos\theta+\sin\theta)^4\mathrm{d}\theta+\int_0^{2\pi}\mathrm{d}\theta\int_0^2 r^3\mathrm{d}r=\frac{16}{3}\int_{-\frac{\pi}{4}}^{\frac{\pi}{4}}\sin^4\left(\theta+\frac{\pi}{4}\right)\mathrm{d}\theta+8\pi$$

$$\xlongequal{\theta+\frac{\pi}{4}=t}\frac{16}{3}\int_0^{\frac{\pi}{2}}\sin^4 t\mathrm{d}t+8\pi=\frac{16}{3}\cdot\frac{3}{4}\cdot\frac{1}{2}\cdot\frac{\pi}{2}+8\pi=9\pi.$$

18.【解析】作出区域D，如图所示.在D中作曲线$y=\frac{1}{x}$及直线$x=\frac{1}{2}$，将D划分成D_1，D_2及D_3三部分(见图)，则



$$|xy-1|=\begin{cases}1-xy, & (x,y)\in D_1,\\ 1-xy, & (x,y)\in D_2,\\ xy-1, & (x,y)\in D_3,\end{cases}$$

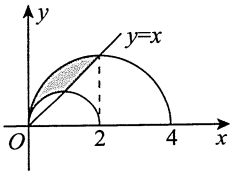
$$\iint_D|xy-1|\,\mathrm{d}\sigma=\iint_{D_1}(1-xy)\mathrm{d}\sigma+\iint_{D_2}(1-xy)\mathrm{d}\sigma+\iint_{D_3}(xy-1)\mathrm{d}\sigma$$

$$=\int_0^{\frac{1}{2}}\mathrm{d}x\int_0^2(1-xy)\mathrm{d}y+\int_{\frac{1}{2}}^2\mathrm{d}x\int_0^{\frac{1}{x}}(1-xy)\mathrm{d}y+\int_{\frac{1}{2}}^2\mathrm{d}x\int_{\frac{1}{x}}^2(xy-1)\mathrm{d}y$$

$$=\int_0^{\frac{1}{2}}(2-2x)\mathrm{d}x+\int_{\frac{1}{2}}^2\frac{1}{2x}\mathrm{d}x+\int_{\frac{1}{2}}^2\left(2x-2+\frac{1}{2x}\right)\mathrm{d}x=\frac{3}{2}+2\ln 2.$$

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19.【解析】积分区域 D 如图所示.



$$\iint_D (x^2+y^2)\mathrm{d}\sigma=\int_{\frac{\pi}{4}}^{\frac{\pi}{2}}\mathrm{d}\theta\int_{2\cos\theta}^{4\cos\theta} r^3\mathrm{d}r=\int_{\frac{\pi}{4}}^{\frac{\pi}{2}}\left(\frac{1}{4}r^4\Big|_{2\cos\theta}^{4\cos\theta}\right)\mathrm{d}\theta$$

$$=60\int_{\frac{\pi}{4}}^{\frac{\pi}{2}}\cos^4\theta\mathrm{d}\theta=60\int_{\frac{\pi}{4}}^{\frac{\pi}{2}}\frac{(1+\cos 2\theta)^2}{4}\mathrm{d}\theta$$

$$=15\int_{\frac{\pi}{4}}^{\frac{\pi}{2}}(1+2\cos 2\theta+\cos^2 2\theta)\mathrm{d}\theta$$

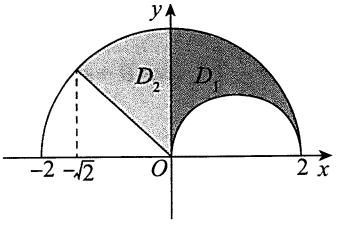
$$=15\left(\frac{1}{4}\pi+\sin 2\theta\Big|_{\frac{\pi}{4}}^{\frac{\pi}{2}}+\int_{\frac{\pi}{4}}^{\frac{\pi}{2}}\frac{1+\cos 4\theta}{2}\mathrm{d}\theta\right)$$

$$=15\left(\frac{1}{4}\pi-1+\frac{1}{2}\cdot\frac{\pi}{4}+\frac{1}{8}\sin 4\theta\Big|_{\frac{\pi}{4}}^{\frac{\pi}{2}}\right)$$

$$=15\left(\frac{1}{4}\pi-1+\frac{\pi}{8}\right)=15\left(\frac{3}{8}\pi-1\right).$$

20.【答案】 $2\pi-\dfrac{16}{9}$

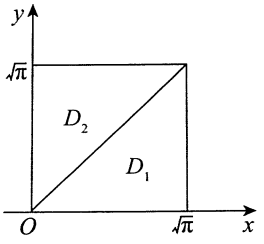
【解析】画出积分区域(见图),用极坐标.



$$\text{原式}=\iint_{D_1\cup D_2} r^2\mathrm{d}r\mathrm{d}\theta=\iint_{D_1} r^2\mathrm{d}r\mathrm{d}\theta+\iint_{D_2} r^2\mathrm{d}r\mathrm{d}\theta$$

$$=\int_0^{\frac{\pi}{2}}\mathrm{d}\theta\int_{2\cos\theta}^{2} r^2\mathrm{d}r+\int_{\frac{\pi}{2}}^{\frac{3\pi}{4}}\mathrm{d}\theta\int_0^2 r^2\mathrm{d}r=2\pi-\frac{16}{9}.$$

21.【解析】 D 是一正方形区域,如图所示.



$$\iint_D \sin\left(\max\left\{x^2,y^2\right\}\right)\mathrm{d}\sigma=\iint_{D_1}\sin x^2\mathrm{d}\sigma+\iint_{D_2}\sin y^2\mathrm{d}\sigma$$

$$=2\iint_{D_1}\sin x^2\mathrm{d}\sigma=2\int_0^{\sqrt{\pi}}\mathrm{d}x\int_0^x\sin x^2\mathrm{d}y$$

$$=2\int_0^{\sqrt{\pi}}x\sin x^2\mathrm{d}x=\int_0^{\sqrt{\pi}}\sin x^2\mathrm{d}(x^2)$$

$$=-\cos x^2\Big|_0^{\sqrt{\pi}}=-(-1-1)=2.$$

22.【答案】 $\dfrac{\pi}{20}$

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【解析】交换积分次序，得

$$\int_0^1 \mathrm{d}y \int_{\sqrt{y}}^1 \sqrt{x^4-y^2}\,\mathrm{d}x = \int_0^1 \mathrm{d}x \int_0^{x^2} \sqrt{x^4-y^2}\,\mathrm{d}y$$

$$= \int_0^1 \mathrm{d}x \int_0^{x^2} \sqrt{(x^2)^2-y^2}\,\mathrm{d}y = \int_0^1 \frac{\pi}{4} x^4 \mathrm{d}x = \frac{\pi}{20}.$$

【注】$\int_0^a \sqrt{a^2-y^2}\,\mathrm{d}y (a>0)$ 是以 a 为半径的圆在第一象限部分的面积，利用定积分的几何意义计算定积分是一种“技巧”.

23. 【答案】$\frac{1}{6}-\frac{1}{3\mathrm{e}}$

【解析】交换积分次序，得

$$\int_0^1 x^2 \mathrm{d}x \int_x^1 \mathrm{e}^{-y^2}\mathrm{d}y = \int_0^1 \mathrm{e}^{-y^2}\mathrm{d}y \int_0^y x^2 \mathrm{d}x$$

$$= \int_0^1 \mathrm{e}^{-y^2} \cdot \frac{y^3}{3}\,\mathrm{d}y = \frac{1}{3}\int_0^1 y^3 \cdot \mathrm{e}^{-y^2}\mathrm{d}y$$

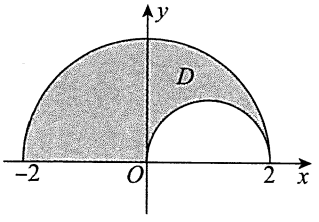
$$= \frac{1}{3} \cdot \left(-\frac{1}{2}\right) \int_0^1 y^2 \mathrm{d}(\mathrm{e}^{-y^2}) \xlongequal{y^2=t} -\frac{1}{6}\int_0^1 t\,\mathrm{d}(\mathrm{e}^{-t})$$

$$= -\frac{1}{6}(t\mathrm{e}^{-t}+\mathrm{e}^{-t})\Big|_0^1 = \frac{1}{6}-\frac{1}{3\mathrm{e}}.$$

24. 【解析】积分区域 D 如图所示，由极坐标表示：

$x^2+y^2=4$，$y\geqslant 0$ 化为 $r=2$，$0\leqslant\theta\leqslant\pi$；

$(x-1)^2+y^2=1$，$y\geqslant 0$ 化为 $r=2\cos\theta$，$0\leqslant\theta\leqslant\frac{\pi}{2}$.



$$\iint_D (xy+y^2)\mathrm{d}\sigma = \int_0^{\frac{\pi}{2}}\mathrm{d}\theta \int_{2\cos\theta}^2 (r^2\sin\theta\cos\theta + r^2\sin^2\theta) r\mathrm{d}r + \int_{\frac{\pi}{2}}^{\pi}\mathrm{d}\theta\int_0^2 (r^2\sin\theta\cos\theta + r^2\sin^2\theta) r\mathrm{d}r$$

$$= \int_0^{\frac{\pi}{2}} 4(1-\cos^4\theta)(\sin\theta\cos\theta+\sin^2\theta)\mathrm{d}\theta + \int_{\frac{\pi}{2}}^{\pi} 4(\sin\theta\cos\theta+\sin^2\theta)\mathrm{d}\theta$$

$$= 4\int_0^{\pi}(\sin\theta\cos\theta+\sin^2\theta)\mathrm{d}\theta - 4\int_0^{\frac{\pi}{2}}\cos^5\theta\sin\theta\,\mathrm{d}\theta - 4\int_0^{\frac{\pi}{2}}\cos^4\theta\sin^2\theta\,\mathrm{d}\theta$$

$$= 0 + 8\int_0^{\frac{\pi}{2}}\sin^2\theta\mathrm{d}\theta + 4\int_0^{\frac{\pi}{2}}\cos^5\theta\mathrm{d}(\cos\theta) - 4\int_0^{\frac{\pi}{2}}\cos^4\theta\mathrm{d}\theta + 4\int_0^{\frac{\pi}{2}}\cos^6\theta\mathrm{d}\theta$$

$$= 8\cdot\frac{1}{2}\cdot\frac{\pi}{2} + \frac{2}{3}\left(\cos^6\frac{\pi}{2}-1\right) - 4\cdot\frac{3}{4}\cdot\frac{1}{2}\cdot\frac{\pi}{2} + 4\cdot\frac{5}{6}\cdot\frac{3}{4}\cdot\frac{1}{2}\cdot\frac{\pi}{2}$$

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$$=\frac{15}{8}\pi-\frac{2}{3}.$$

【注】计算$\int_0^{\frac{\pi}{2}}\sin^n x\mathrm{d}x$与$\int_0^{\frac{\pi}{2}}\cos^n x\mathrm{d}x$用到华里士公式，请考生切记此公式.

25.【答案】$\frac{5\pi}{2}$

【解析】方法一 令$x=1+r\cos\theta$，$y=1+r\sin\theta$，则

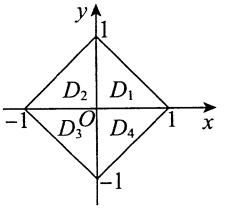
$$\begin{aligned}\iint_D(x^2+y^2)\mathrm{d}\sigma&=\int_0^{2\pi}\mathrm{d}\theta\int_0^1(r^2+2r\cos\theta+2r\sin\theta+2)r\mathrm{d}r\\&=\int_0^{2\pi}\mathrm{d}\theta\int_0^1(r^3+2r^2\cos\theta+2r^2\sin\theta+2r)\mathrm{d}r\\&=\int_0^{2\pi}\left(\frac{5}{4}+\frac{2}{3}\cos\theta+\frac{2}{3}\sin\theta\right)\mathrm{d}\theta\\&=\left(\frac{5}{4}\theta+\frac{2}{3}\sin\theta-\frac{2}{3}\cos\theta\right)\Big|_0^{2\pi}=\frac{5\pi}{2}.\end{aligned}$$

方法二 作变换$x=u+1$，$y=v+1$，在此变换下，与D对应的闭区域为$D'=\{(u,v)\mid u^2+v^2\leqslant 1\}$，则有$J(u,v)=\frac{\partial(x,y)}{\partial(u,v)}=1$，从而有

$$\begin{aligned}\iint_D(x^2+y^2)\mathrm{d}\sigma&=\iint_{D'}\left[(u+1)^2+(v+1)^2\right]|J(u,v)|\,\mathrm{d}\sigma\\&=\iint_{D'}(u^2+v^2+2)\mathrm{d}\sigma\\&=\iint_{D'}(u^2+v^2)\mathrm{d}\sigma+\iint_{D'}2\mathrm{d}\sigma\\&=\int_0^{2\pi}\mathrm{d}\theta\int_0^1 r^3\mathrm{d}r+2\pi=\frac{5\pi}{2}.\end{aligned}$$

26.【解析】将D分成第一、二、三、四象限中的4块，分别记为D_1，D_2，D_3，D_4(见图)，用极坐标，先对D_1进行积分：

$$\begin{aligned}\iint_{D_1}\mathrm{e}^{\frac{|y|}{|x|+|y|}}\mathrm{d}\sigma&=\int_0^{\frac{\pi}{2}}\mathrm{d}\theta\int_0^{\frac{1}{\sin\theta+\cos\theta}}\mathrm{e}^{\frac{\sin\theta}{\sin\theta+\cos\theta}}r\mathrm{d}r\\&=\frac{1}{2}\int_0^{\frac{\pi}{2}}\mathrm{e}^{\frac{\sin\theta}{\sin\theta+\cos\theta}}\cdot\frac{1}{(\sin\theta+\cos\theta)^2}\mathrm{d}\theta\end{aligned}$$



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$$=\frac{1}{2}\int_0^{\frac{\pi}{2}}e^{\frac{\sin\theta}{\sin\theta+\cos\theta}}\mathrm{d}\left(\frac{\sin\theta}{\sin\theta+\cos\theta}\right)$$

$$=\frac{1}{2}e^{\frac{\sin\theta}{\sin\theta+\cos\theta}}\Bigg|_0^{\frac{\pi}{2}}=\frac{1}{2}(e-1).$$

根据对称性，被积函数关于 x，y 均为偶函数，且积分区域关于 y 轴，x 轴对称，故

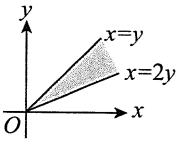
$$\iint_D e^{\frac{|y|}{|x|+|y|}}d\sigma=4\iint_{D_1}e^{\frac{|y|}{|x|+|y|}}d\sigma=4\cdot\frac{1}{2}(e-1)=2(e-1).$$

27.【答案】(D)

【解析】积分区域如图所示.

$$\int_0^{+\infty}dy\int_y^{2y}e^{-x^2}dx=\int_0^{+\infty}dx\int_{\frac{x}{2}}^{x}e^{-x^2}dy=\int_0^{+\infty}\frac{x}{2}e^{-x^2}dx$$

$$=-\frac{1}{4}\int_0^{+\infty}e^{-x^2}d(-x^2)=-\frac{1}{4}\cdot e^{-x^2}\Bigg|_0^{+\infty}=\frac{1}{4}.$$



28.【答案】$-\pi\left(e^{\pi}-e^{\frac{1}{\pi}}\right)$

【解析】当 $t>1$ 时，设区域 $D=\left\{(x,y)\,\middle|\,1\leqslant x\leqslant t^2,\sqrt{x}\leqslant y\leqslant t\right\}$，从而

$$f(t)=-\int_1^{t^2}dx\int_{\sqrt{x}}^{t}e^{\frac{x}{y}}dy=-\iint_D e^{\frac{x}{y}}dxdy,$$

将二重积分写为先对 x，后对 y 的累次积分，有 $f(t)=-\int_1^t dy\int_1^{y^2}e^{\frac{x}{y}}dx=-\int_1^t y\left(e^{y}-e^{\frac{1}{y}}\right)dy$，所以

$$f'(t)=-t\left(e^{t}-e^{\frac{1}{t}}\right),$$

从而 $f'(\pi)=-\pi\left(e^{\pi}-e^{\frac{1}{\pi}}\right)$.

【注】由于所求为 $f'(\pi)$，$\pi>1$，故只研究 $t>1$ 即可.

29.【答案】4π

【解析】方法一　在极坐标系中，将 $(x-1)^2+(y-1)^2=2$ 化为 $r=2(\cos\theta+\sin\theta)=2\sqrt{2}\sin\left(\frac{\pi}{4}+\theta\right)$.

$$\iint_D(x+y)d\sigma=\int_{-\frac{\pi}{4}}^{\frac{3\pi}{4}}d\theta\int_0^{2\sqrt{2}\sin\left(\frac{\pi}{4}+\theta\right)}r^2(\cos\theta+\sin\theta)dr$$

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$$=\int_{-\frac{\pi}{4}}^{\frac{3\pi}{4}}\mathrm{d}\theta\int_0^{2\sqrt{2}\sin\left(\frac{\pi}{4}+\theta\right)}\sqrt{2}r^2\sin\left(\frac{\pi}{4}+\theta\right)\mathrm{d}r$$

$$=\frac{2^5}{3}\int_{-\frac{\pi}{4}}^{\frac{3\pi}{4}}\sin^4\left(\frac{\pi}{4}+\theta\right)\mathrm{d}\theta=\frac{2^5}{3}\int_0^{\pi}\sin^4t\mathrm{d}t$$

$$=\frac{64}{3}\int_0^{\frac{\pi}{2}}\sin^4t\mathrm{d}t=\frac{64}{3}\cdot\frac{3}{4}\cdot\frac{1}{2}\cdot\frac{\pi}{2}=4\pi.$$

方法二 用直角坐标.

$$\iint_D(x+y)\mathrm{d}\sigma=\iint_D\left[(x-1)+(y-1)+2\right]\mathrm{d}\sigma$$

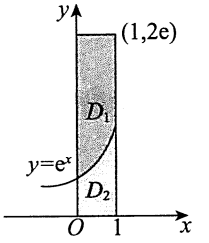
$$=\iint_D(x-1)\mathrm{d}\sigma+\iint_D(y-1)\mathrm{d}\sigma+2\iint_D\mathrm{d}\sigma.$$

$$\iint_D(x-1)\mathrm{d}\sigma=\int_{1-\sqrt{2}}^{1+\sqrt{2}}\mathrm{d}x\int_{1-\sqrt{2-(x-1)^2}}^{1+\sqrt{2-(x-1)^2}}(x-1)\mathrm{d}y$$

$$=\int_{1-\sqrt{2}}^{1+\sqrt{2}}2(x-1)\sqrt{2-(x-1)^2}\mathrm{d}x\xlongequal{x-1=t}\int_{-\sqrt{2}}^{\sqrt{2}}2t\sqrt{2-t^2}\mathrm{d}t=0.$$

同理，$\iint_D(y-1)\mathrm{d}\sigma=0$，而$2\iint_D\mathrm{d}\sigma=2\cdot2\pi=4\pi$，所以原式$=4\pi$.

30. 【解析】如图所示，将D分成$D_1\cup D_2$，D_1与D_2除边界之外无公共部分.



$$\iint_D x\left|y-\mathrm{e}^x\right|\mathrm{d}\sigma=\iint_{D_1}x\left|y-\mathrm{e}^x\right|\mathrm{d}\sigma+\iint_{D_2}x\left|y-\mathrm{e}^x\right|\mathrm{d}\sigma$$

$$=\int_0^1\mathrm{d}x\int_{\mathrm{e}^x}^{2\mathrm{e}}x(y-\mathrm{e}^x)\mathrm{d}y+\int_0^1\mathrm{d}x\int_0^{\mathrm{e}^x}x(\mathrm{e}^x-y)\mathrm{d}y$$

$$=\int_0^1\left[\left(\frac{xy^2}{2}-yx\mathrm{e}^x\right)\Big|_{y=\mathrm{e}^x}^{y=2\mathrm{e}}\right]\mathrm{d}x+\int_0^1\left[\left(yx\mathrm{e}^x-\frac{xy^2}{2}\right)\Big|_{y=0}^{y=\mathrm{e}^x}\right]\mathrm{d}x$$

$$=\int_0^1\left(2\mathrm{e}^2x-2\mathrm{e}x\mathrm{e}^x+\frac{1}{2}x\mathrm{e}^{2x}\right)\mathrm{d}x+\int_0^1\frac{x}{2}\mathrm{e}^{2x}\mathrm{d}x$$

$$=\frac{5}{4}\mathrm{e}^2-2\mathrm{e}+\frac{1}{4}.$$

31. 【解析】如图所示，曲线$x^2+y^2=1$将D分成三块，中间一块记为D_3，左、右两块分别记为D_1与D_2，则

$$f(x,y)=\max\left\{\sqrt{x^2+y^2},1\right\}$$

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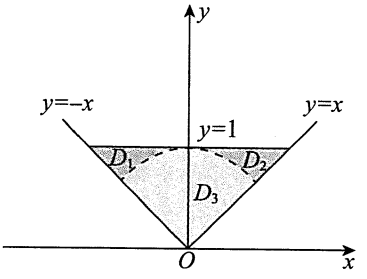
$$=\begin{cases}\sqrt{x^2+y^2}, & (x,y)\in D_1\cup D_2,\\ 1, & (x,y)\in D_3,\end{cases}$$

$$\iint_D f(x,y)\mathrm{d}\sigma=\iint_{D_1\cup D_2}\sqrt{x^2+y^2}\mathrm{d}\sigma+\iint_{D_3}\mathrm{d}\sigma$$

$$=\int_{\frac{\pi}{4}}^{\frac{3\pi}{4}}\mathrm{d}\theta\int_1^{\frac{1}{\sin\theta}}r^2\mathrm{d}r+\frac{\pi}{4}$$

$$=\frac{1}{3}\int_{\frac{\pi}{4}}^{\frac{3\pi}{4}}(\csc^3\theta-1)\mathrm{d}\theta+\frac{\pi}{4}$$

$$=\frac{1}{3}\int_{\frac{\pi}{4}}^{\frac{3\pi}{4}}\csc^3\theta\mathrm{d}\theta+\frac{\pi}{12}.$$



而　$\int_{\frac{\pi}{4}}^{\frac{3\pi}{4}}\csc^3\theta\mathrm{d}\theta=\int_{\frac{\pi}{4}}^{\frac{3\pi}{4}}\csc\theta\csc^2\theta\mathrm{d}\theta$

$$=-\int_{\frac{\pi}{4}}^{\frac{3\pi}{4}}\csc\theta\mathrm{d}(\cot\theta)=-\left(\csc\theta\cot\theta\Big|_{\frac{\pi}{4}}^{\frac{3\pi}{4}}+\int_{\frac{\pi}{4}}^{\frac{3\pi}{4}}\cot^2\theta\csc\theta\mathrm{d}\theta\right)$$

$$=-\left[-\sqrt{2}-\sqrt{2}+\int_{\frac{\pi}{4}}^{\frac{3\pi}{4}}(\csc^2\theta-1)\csc\theta\mathrm{d}\theta\right],$$

所以

$$\int_{\frac{\pi}{4}}^{\frac{3\pi}{4}}\csc^3\theta\mathrm{d}\theta=\sqrt{2}+\frac{1}{2}\int_{\frac{\pi}{4}}^{\frac{3\pi}{4}}\csc\theta\mathrm{d}\theta$$

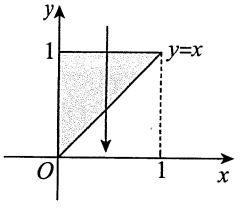
$$=\sqrt{2}+\frac{1}{2}(\ln|\csc\theta-\cot\theta|)\Big|_{\frac{\pi}{4}}^{\frac{3\pi}{4}}$$

$$=\sqrt{2}+\frac{1}{2}\left[\ln(\sqrt{2}+1)-\ln(\sqrt{2}-1)\right]$$

$$=\sqrt{2}+\ln(\sqrt{2}+1).$$

故原式 $=\frac{\sqrt{2}}{3}+\frac{1}{3}\ln(\sqrt{2}+1)+\frac{\pi}{12}$.

32.【解析】积分区域如图所示，则



$$\text{原式}=-\int_0^1\mathrm{d}x\int_x^1(\mathrm{e}^{-y^2}+\mathrm{e}^y\sin y)\mathrm{d}y$$

$$=-\int_0^1\mathrm{d}y\int_0^y(\mathrm{e}^{-y^2}+\mathrm{e}^y\sin y)\mathrm{d}x$$

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$$=-\int_0^1 ye^{-y^2}dy-\int_0^1 ye^y\sin ydy$$

$$=\frac{1}{2}e^{-y^2}\Big|_0^1-\int_0^1 ye^y\sin ydy$$

$$=\frac{1}{2}e^{-1}-\frac{1}{2}-\int_0^1 ye^y\sin ydy.$$

其中，

$$\int_0^1 ye^y\sin ydy=\int_0^1 y\sin yd(e^y)$$

$$=y\sin y\cdot e^y\Big|_0^1-\int_0^1 e^y(\sin y+y\cos y)dy$$

$$=e\sin 1-\int_0^1 e^y\sin ydy-\int_0^1 y\cos yd(e^y)$$

$$=e\sin 1-\int_0^1 e^y\sin ydy-y\cos y\cdot e^y\Big|_0^1+\int_0^1 e^y(\cos y-y\sin y)dy$$

$$=e(\sin 1-\cos 1)+e^y\cos y\Big|_0^1-\int_0^1 ye^y\sin ydy$$

$$=e\sin 1-1-\int_0^1 ye^y\sin ydy,$$

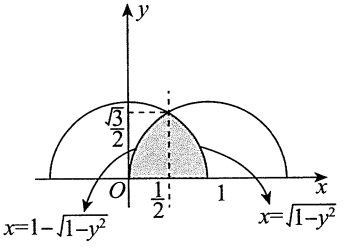
则
$$\int_0^1 ye^y\sin ydy=\frac{1}{2}(e\sin 1-1).$$

故
$$原式=\frac{1}{2}e^{-1}-\frac{1}{2}-\frac{1}{2}(e\sin 1-1)$$

$$=\frac{e^{-1}-e\sin 1}{2}.$$

33.【解析】转化为直角坐标系再计算.

由 $r\leqslant 1$，知 $x^2+y^2\leqslant 1$；由 $r\leqslant 2\cos\theta$，知 $(x-1)^2+y^2\leqslant 1$；由 $\sin\theta\geqslant 0$，知 $y\geqslant 0$.积分区域 D_{xy} 如图所示.



$$原式=\iint_D r\cos\theta(1+r\sin\theta)rdrd\theta$$

$$=\iint_{D_{xy}} x(1+y)dxdy$$

$$=\iint_{D_{xy}} xdxdy+\iint_{D_{xy}} xydxdy$$

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$$=\iint_{D_{xy}}\left(x-\frac{1}{2}\right)\mathrm{d}x\mathrm{d}y+\frac{1}{2}S_{D_{xy}}+\iint_{D_{xy}}xy\mathrm{d}x\mathrm{d}y$$

$$=0+\frac{1}{2}\times 2\int_0^{\frac{\sqrt{3}}{2}}\mathrm{d}y\int_{1-\sqrt{1-y^2}}^{\frac{1}{2}}\mathrm{d}x+\int_0^{\frac{\sqrt{3}}{2}}y\mathrm{d}y\int_{1-\sqrt{1-y^2}}^{\sqrt{1-y^2}}x\mathrm{d}x$$

$$=\int_0^{\frac{\sqrt{3}}{2}}\sqrt{1-y^2}\,\mathrm{d}y-\frac{\sqrt{3}}{4}+\int_0^{\frac{\sqrt{3}}{2}}\left(y\sqrt{1-y^2}-\frac{1}{2}y\right)\mathrm{d}y$$

$$=\frac{\pi}{6}+\frac{\sqrt{3}}{8}-\frac{\sqrt{3}}{4}+\frac{5}{48}$$

$$=\frac{\pi}{6}-\frac{\sqrt{3}}{8}+\frac{5}{48}.$$

34.【解析】令 $a=\iint_D\frac{(2x^2+1)f(x,y)}{x^2+y^2+1}\mathrm{d}x\mathrm{d}y$，则 $f(x,y)=\mathrm{e}^{x^2+y^2}-a=f(y,x)$．由轮换对称性，得

$$a=\iint_D\frac{(2y^2+1)f(x,y)}{y^2+x^2+1}\mathrm{d}x\mathrm{d}y$$

$$=\frac{1}{2}\iint_D 2f(x,y)\mathrm{d}x\mathrm{d}y$$

$$=\iint_D f(x,y)\mathrm{d}x\mathrm{d}y,$$

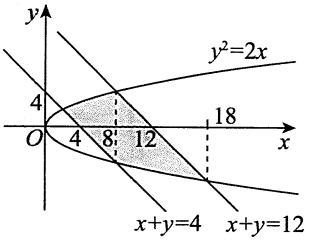
于是

$$a=\iint_D(\mathrm{e}^{x^2+y^2}-a)\mathrm{d}x\mathrm{d}y$$

$$=\int_0^{2\pi}\mathrm{d}\theta\int_0^1(\mathrm{e}^{r^2}-a)r\mathrm{d}r$$

$$=(\mathrm{e}-a-1)\pi,$$

解得 $a=\frac{(\mathrm{e}-1)\pi}{\pi+1}$，故 $\iint_D f(x,y)\mathrm{d}x\mathrm{d}y=\frac{(\mathrm{e}-1)\pi}{\pi+1}$．



35.【解析】方法一　由题可知，积分区域如图所示．

易求得 $y^2=2x$ 与 $x+y=4$ 的交点为 $(2,2)$，$(8,-4)$；$y^2=2x$ 与 $x+y=12$ 的交点为 $(8,4)$，$(18,-6)$．故

$$\iint_D\frac{2x+y}{x^{\frac{3}{2}}}\mathrm{d}x\mathrm{d}y=\int_2^8\mathrm{d}x\int_{4-x}^{\sqrt{2x}}\frac{2x+y}{x^{\frac{3}{2}}}\mathrm{d}y+\int_8^{18}\mathrm{d}x\int_{-\sqrt{2x}}^{12-x}\frac{2x+y}{x^{\frac{3}{2}}}\mathrm{d}y,$$

其中，

$$\int_2^8\mathrm{d}x\int_{4-x}^{\sqrt{2x}}\frac{2x+y}{x^{\frac{3}{2}}}\mathrm{d}y=\int_2^8\left(2\sqrt{2}-3x^{-\frac{1}{2}}+\frac{3}{2}x^{\frac{1}{2}}-8x^{-\frac{3}{2}}\right)\mathrm{d}x$$

$$=12\sqrt{2}-6\sqrt{2}+14\sqrt{2}-4\sqrt{2}=16\sqrt{2},$$

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$$\int_8^{18}\mathrm{d}x\int_{-\sqrt{2x}}^{12-x}\frac{2x+y}{x^{\frac{3}{2}}}\mathrm{d}y=\int_8^{18}\left(2\sqrt{2}-\frac{3}{2}x^{\frac{1}{2}}+11x^{-\frac{1}{2}}+72x^{-\frac{3}{2}}\right)\mathrm{d}x$$

$$=20\sqrt{2}-38\sqrt{2}+22\sqrt{2}+12\sqrt{2}=16\sqrt{2}.$$

故 $$\iint_D\frac{2x+y}{x^{\frac{3}{2}}}\mathrm{d}x\mathrm{d}y=16\sqrt{2}+16\sqrt{2}=32\sqrt{2}.$$

方法二 令$\begin{cases}y=\xi\sqrt{x},\\x+y=\eta,\end{cases}$从中解出$\xi$，$\eta$，即得变量代换$\begin{cases}\xi=\dfrac{y}{\sqrt{x}},\\\eta=x+y.\end{cases}$

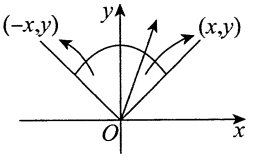
在这个变换下，xOy平面上的区域D变成$\xi O\eta$平面上的矩形区域D'，其中$D'=\{(\xi,\eta)\mid-\sqrt{2}\leqslant\xi\leqslant\sqrt{2}，4\leqslant\eta\leqslant12\}$，其雅可比行列式为

$$J=\frac{\partial(x,y)}{\partial(\xi,\eta)}=\frac{1}{\dfrac{\partial(\xi,\eta)}{\partial(x,y)}}=\frac{1}{\begin{vmatrix}\dfrac{\partial\xi}{\partial x}&\dfrac{\partial\xi}{\partial y}\\[2mm]\dfrac{\partial\eta}{\partial x}&\dfrac{\partial\eta}{\partial y}\end{vmatrix}}=\frac{1}{\begin{vmatrix}-\dfrac{y}{2x^{\frac{3}{2}}}&\dfrac{1}{x^{\frac{1}{2}}}\\[2mm]1&1\end{vmatrix}}=\frac{1}{-\dfrac{2x+y}{2x^{\frac{3}{2}}}}=-\frac{2x^{\frac{3}{2}}}{2x+y},$$

于是 $$\iint_D\frac{2x+y}{x^{\frac{3}{2}}}\mathrm{d}x\mathrm{d}y=\iint_{D'}\frac{2x+y}{x^{\frac{3}{2}}}\left|\frac{\partial(x,y)}{\partial(\xi,\eta)}\right|\mathrm{d}\xi\mathrm{d}\eta$$

$$=\iint_{D'}2\mathrm{d}\xi\mathrm{d}\eta=32\sqrt{2}.$$

36. 【解析】将$x^2+y^2=\sqrt{x^2+y^2}+\dfrac{y}{2}$进行极坐标变换得$r^2=r+\dfrac{1}{2}r\sin\theta$，即$r=1+\dfrac{\sin\theta}{2}$.由于平面区域$D$关于$y$轴对称(见图)，故$\iint_D\dfrac{x}{\sqrt{x^2+y^2}}\mathrm{d}x\mathrm{d}y=0$.又



$$\iint_D\frac{y}{\sqrt{x^2+y^2}}\mathrm{d}x\mathrm{d}y=2\int_{\frac{\pi}{4}}^{\frac{\pi}{2}}\mathrm{d}\theta\int_0^{1+\frac{\sin\theta}{2}}\frac{r\sin\theta}{r}r\mathrm{d}r$$

$$=\int_{\frac{\pi}{4}}^{\frac{\pi}{2}}\sin\theta\cdot\left(1+\frac{\sin\theta}{2}\right)^2\mathrm{d}\theta$$

$$=\int_{\frac{\pi}{4}}^{\frac{\pi}{2}}\left(\sin\theta+\sin^2\theta+\frac{1}{4}\sin^3\theta\right)\mathrm{d}\theta$$

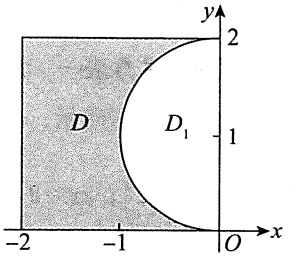
$$=-\cos\theta\Big|_{\frac{\pi}{4}}^{\frac{\pi}{2}}+\frac{1}{2}\cdot\frac{\pi}{4}-\frac{1}{4}\sin2\theta\Big|_{\frac{\pi}{4}}^{\frac{\pi}{2}}-\frac{1}{4}\left(\cos\theta\Big|_{\frac{\pi}{4}}^{\frac{\pi}{2}}-\frac{1}{3}\cos^3\theta\Big|_{\frac{\pi}{4}}^{\frac{\pi}{2}}\right)$$

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$$=\frac{29}{48}\sqrt{2}+\frac{\pi}{8}+\frac{1}{4},$$

所以 $\iint\limits_D \frac{x+y}{\sqrt{x^2+y^2}}\mathrm{d}x\mathrm{d}y=\frac{29}{48}\sqrt{2}+\frac{\pi}{8}+\frac{1}{4}$.

37.【解析】区域 D 和 D_1 如图所示.



$$I=\iint\limits_{D+D_1}x^2y^2\mathrm{d}x\mathrm{d}y-\iint\limits_{D_1}x^2y^2\mathrm{d}x\mathrm{d}y,$$

其中
$$\iint\limits_{D+D_1}x^2y^2\mathrm{d}x\mathrm{d}y=\int_{-2}^{0}\mathrm{d}x\int_{0}^{2}x^2y^2\mathrm{d}y$$
$$=\int_{-2}^{0}\frac{8}{3}x^2\mathrm{d}x=\frac{64}{9}.$$

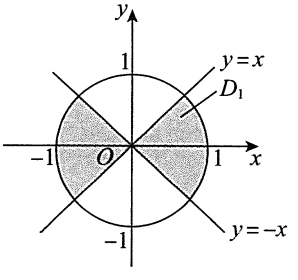
对于 $\iint\limits_{D_1}x^2y^2\mathrm{d}x\mathrm{d}y$，可采用极坐标，得

$$\iint\limits_{D_1}x^2y^2\mathrm{d}x\mathrm{d}y=\iint\limits_{D_1}r^5\sin^2\theta\cos^2\theta\mathrm{d}r\mathrm{d}\theta=\int_{\frac{\pi}{2}}^{\pi}\mathrm{d}\theta\int_{0}^{2\sin\theta}r^5\sin^2\theta\cos^2\theta\mathrm{d}r$$
$$=\frac{32}{3}\int_{\frac{\pi}{2}}^{\pi}\sin^8\theta(1-\sin^2\theta)\mathrm{d}\theta=\frac{32}{3}\left(\int_0^{\frac{\pi}{2}}\cos^8u\mathrm{d}u-\int_0^{\frac{\pi}{2}}\cos^{10}u\mathrm{d}u\right)$$
$$\overset{\text{华里士公式}}{=\!=\!=}\frac{32}{3}\left(\frac{7}{8}\times\frac{5}{6}\times\frac{3}{4}\times\frac{1}{2}\times\frac{\pi}{2}-\frac{9}{10}\times\frac{7}{8}\times\frac{5}{6}\times\frac{3}{4}\times\frac{1}{2}\times\frac{\pi}{2}\right)$$
$$=\frac{7}{48}\pi.$$

从而
$$I=\frac{64}{9}-\frac{7}{48}\pi.$$

38.【解析】
$$I=\iint\limits_D\frac{|xy|}{\sqrt{x^2+y^2}+x^2-y^2}\mathrm{d}\sigma=4\iint\limits_{D_1}\frac{xy}{\sqrt{x^2+y^2}+x^2-y^2}\mathrm{d}\sigma,$$

其中 $D_1=\left\{(x,y)\middle|x^2+y^2\leqslant1,0\leqslant y\leqslant x\right\}$，如图所示.



$$\iint\limits_{D_1}\frac{xy}{\sqrt{x^2+y^2}+x^2-y^2}\mathrm{d}\sigma=\frac{1}{2}\int_0^1r\mathrm{d}r\int_0^{\frac{\pi}{4}}\frac{r^2\sin2\theta}{r+r^2\cos2\theta}\mathrm{d}\theta$$
$$=-\frac{1}{4}\int_0^1r\mathrm{d}r\int_0^{\frac{\pi}{4}}\frac{\mathrm{d}(1+r\cos2\theta)}{1+r\cos2\theta}$$
$$=\frac{1}{4}\int_0^1r\ln(1+r)\mathrm{d}r$$

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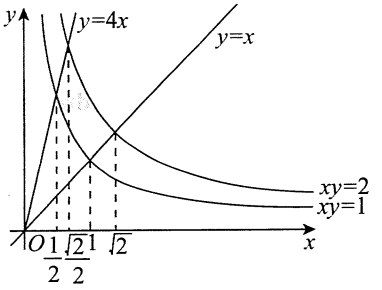
$$=\frac{1}{8}\left[r^2\ln(1+r)\Big|_0^1-\int_0^1\frac{r^2}{1+r}\,\mathrm{d}r\right]$$

$$=\frac{1}{8}\left[\ln 2-\int_0^1(r-1)\mathrm{d}r-\int_0^1\frac{1}{1+r}\,\mathrm{d}r\right]=\frac{1}{16},$$

故 $I=4\times\frac{1}{16}=\frac{1}{4}$.

39.【解析】方法一 常规做法.

积分区域如图所示，$xy=1$ 与 $y=x$ 及 $y=4x$ 的交点坐标分别为 $(1,1)$，$\left(\frac{1}{2},2\right)$；$xy=2$ 与 $y=x$ 及 $y=4x$ 的交点坐标分别为 $(\sqrt{2},\sqrt{2})$，$\left(\frac{\sqrt{2}}{2},2\sqrt{2}\right)$. 故



$$\iint_D\ln(xy)\,\mathrm{d}x\mathrm{d}y=\int_{\frac{1}{2}}^{\frac{\sqrt{2}}{2}}\mathrm{d}x\int_{\frac{1}{x}}^{4x}\ln(xy)\mathrm{d}y+\int_{\frac{\sqrt{2}}{2}}^{1}\mathrm{d}x\int_{\frac{1}{x}}^{\frac{2}{x}}\ln(xy)\mathrm{d}y+\int_1^{\sqrt{2}}\mathrm{d}x\int_x^{\frac{2}{x}}\ln(xy)\mathrm{d}y$$
$$=I_1+I_2+I_3,$$

其中 $\int\ln(xy)\mathrm{d}y=y\ln(xy)-y+C$，则

$$\int_{\frac{1}{x}}^{4x}\ln(xy)\mathrm{d}y=4(2\ln 2-1)x+8x\ln x+\frac{1}{x},$$

$$\int_{\frac{1}{x}}^{\frac{2}{x}}\ln(xy)\mathrm{d}y=\frac{2\ln 2-1}{x},$$

$$\int_x^{\frac{2}{x}}\ln(xy)\mathrm{d}y=\frac{2\ln 2-2}{x}-2x\ln x+x,$$

故

$$I_1=\int_{\frac{1}{2}}^{\frac{\sqrt{2}}{2}}\left[4(2\ln 2-1)x+8x\ln x+\frac{1}{x}\right]\mathrm{d}x$$

$$=\left[2(2\ln 2-1)x^2+4x^2\ln x-2x^2+\ln x\right]\Big|_{\frac{1}{2}}^{\frac{\sqrt{2}}{2}}$$

$$=\frac{3}{2}\ln 2-1,$$

$$I_2=\int_{\frac{\sqrt{2}}{2}}^{1}\frac{2\ln 2-1}{x}\mathrm{d}x=(2\ln 2-1)\ln x\Big|_{\frac{\sqrt{2}}{2}}^{1}=\ln^2 2-\frac{1}{2}\ln 2,$$

$$I_3=\int_1^{\sqrt{2}}\left(\frac{2\ln 2-2}{x}-2x\ln x+x\right)\mathrm{d}x$$

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$$=\left[(2\ln 2-2)\ln x-x^2\ln x+x^2\right]\Big|_1^{\sqrt{2}}$$

$$=\ln^2 2-2\ln 2+1,$$

可得 $$\iint_D \ln(xy)\,\mathrm{d}x\mathrm{d}y=\ln 2(2\ln 2-1).$$

方法二 $D=\left\{(x,y)\middle|1\leqslant xy\leqslant 2,1\leqslant\frac{y}{x}\leqslant 4,x>0\right\}$，令 $xy=u$，$\frac{y}{x}=v$，则 $x=\sqrt{\frac{u}{v}}$，$y=\sqrt{uv}$ $(u>0,v>0)$，

$$J=\begin{vmatrix}\frac{1}{2\sqrt{uv}} & -\frac{1}{2}\sqrt{\frac{u}{v^3}}\\ \frac{1}{2}\sqrt{\frac{v}{u}} & \frac{1}{2}\sqrt{\frac{u}{v}}\end{vmatrix}=\frac{1}{2v},$$

$$D'=\left\{(u,v)\middle|1\leqslant u\leqslant 2,1\leqslant v\leqslant 4\right\},$$

所以

$$\iint_D \ln(xy)\,\mathrm{d}x\mathrm{d}y=\iint_{D'}\ln u\cdot\frac{1}{2v}\,\mathrm{d}u\mathrm{d}v=\int_1^2\ln u\,\mathrm{d}u\int_1^4\frac{\mathrm{d}v}{2v}=\ln 2(2\ln 2-1).$$

40.【解析】 $2\int_{-\frac{1}{2}}^{1}\left(\int_0^x \mathrm{e}^{-y^2}\mathrm{d}y\right)\mathrm{d}x=2\left(-\int_{-\frac{1}{2}}^{0}\mathrm{d}x\int_x^0\mathrm{e}^{-y^2}\mathrm{d}y+\int_0^1\mathrm{d}x\int_0^x\mathrm{e}^{-y^2}\mathrm{d}y\right)$

$$=2\left(-\iint\limits_{-\frac{1}{2}\leqslant x\leqslant y\leqslant 0}\mathrm{e}^{-y^2}\mathrm{d}x\mathrm{d}y+\iint\limits_{0\leqslant y\leqslant x\leqslant 1}\mathrm{e}^{-y^2}\mathrm{d}x\mathrm{d}y\right)$$

$$=2\left(-\int_{-\frac{1}{2}}^{0}\mathrm{d}y\int_{-\frac{1}{2}}^{y}\mathrm{e}^{-y^2}\mathrm{d}x+\int_0^1\mathrm{d}y\int_y^1\mathrm{e}^{-y^2}\mathrm{d}x\right)$$

$$=2\left[\int_0^{-\frac{1}{2}}\left(y+\frac{1}{2}\right)\mathrm{e}^{-y^2}\mathrm{d}y+\int_0^1(1-y)\mathrm{e}^{-y^2}\mathrm{d}y\right]$$

$$=-\mathrm{e}^{-y^2}\Big|_0^{-\frac{1}{2}}+\int_0^{-\frac{1}{2}}\mathrm{e}^{-y^2}\mathrm{d}y+2\int_0^1\mathrm{e}^{-y^2}\mathrm{d}y+\mathrm{e}^{-y^2}\Big|_0^1$$

$$=2\int_0^1\mathrm{e}^{-y^2}\mathrm{d}y-\int_0^{\frac{1}{2}}\mathrm{e}^{-y^2}\mathrm{d}y+\frac{1}{\mathrm{e}}-\frac{1}{\sqrt[4]{\mathrm{e}}}$$

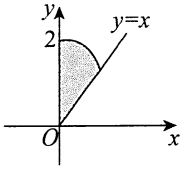
$$=2a-b+\frac{1}{\mathrm{e}}-\frac{1}{\sqrt[4]{\mathrm{e}}},$$

所以 $m=2$，$n=1$.

41.【解析】 (1) $2x^2+y^2=2\sqrt{x^2+y^2}$ 在极坐标系下为 $r(1+\cos^2\theta)=2$，直角坐标系下的积分区域 D 如图所示，故

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$$\iint_D \frac{1}{\sqrt{1-x^2}}\,d\sigma=\int_{\frac{\pi}{4}}^{\frac{\pi}{2}}d\theta\int_0^{\frac{2}{1+\cos^2\theta}}\frac{1}{\sqrt{1-r^2\cos^2\theta}}\,rdr$$



$$=2\int_{\frac{\pi}{4}}^{\frac{\pi}{2}}\frac{1}{1+\cos^2\theta}\,d\theta$$

$$=2\int_{\frac{\pi}{4}}^{\frac{\pi}{2}}\frac{\sec^2\theta}{\sec^2\theta+1}\,d\theta=2\int_{\frac{\pi}{4}}^{\frac{\pi}{2}}\frac{d(\tan\theta)}{2+\tan^2\theta}$$

$$\xlongequal{u=\tan\theta}2\int_1^{+\infty}\frac{du}{2+u^2}=\sqrt{2}\arctan\frac{u}{\sqrt{2}}\Big|_1^{+\infty}$$

$$=\sqrt{2}\cdot\left(\frac{\pi}{2}-\arctan\frac{\sqrt{2}}{2}\right)$$

$$=\frac{\sqrt{2}}{2}\pi-\sqrt{2}\arctan\frac{\sqrt{2}}{2}.$$

(2) 由积分保号性可得

$$a=\iint_D\frac{f(x,y)}{\sqrt{1-x^2}}\,d\sigma=\left|\iint_D\frac{f(x,y)}{\sqrt{1-x^2}}\,d\sigma\right|\leqslant\iint_D\frac{|f(x,y)|}{\sqrt{1-x^2}}\,d\sigma$$

$$\xlongequal{(*)}|f(\xi,\eta)|\iint_D\frac{1}{\sqrt{1-x^2}}\,d\sigma$$

$$=\left(\frac{\sqrt{2}}{2}\pi-\sqrt{2}\arctan\frac{\sqrt{2}}{2}\right)|f(\xi,\eta)|,$$

其中 $(\xi,\eta)\in D$，即存在 $(\xi,\eta)\in D$，使得 $a\leqslant\left(\frac{\sqrt{2}}{2}\pi-\sqrt{2}\arctan\frac{\sqrt{2}}{2}\right)|f(\xi,\eta)|$，即

$$|f(\xi,\eta)|\geqslant\frac{a}{\frac{\sqrt{2}}{2}\pi-\sqrt{2}\arctan\frac{\sqrt{2}}{2}}\geqslant\frac{a}{\frac{\sqrt{2}}{2}\pi}=\frac{\sqrt{2}}{\pi}a.$$

【注】(*)处利用了二重积分的中值定理.

设 $f(x,y)$ 和 $g(x,y)$ 都在闭区域 D 上连续，且 $g(x,y)$ 在 D 上不变号，则至少存在一点 $(\xi,\eta)\in D$，使得 $\iint_D f(x,y)\cdot g(x,y)dxdy=f(\xi,\eta)\iint_D g(x,y)dxdy$.

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